

Share Your Knowledge

If you are an expert in your field of electronics, why not share your knowledge with the world?



Electronics For You invites electronics professionals and industry experts to write articles on their area of expertise and interest from different perspectives

Some of the topics we want you to write about are:

- Photonics: Optics & Optoelectronics
- Automation
- Power Electronics
- Components
- Telecom



So if you have an interesting topic or original article to share, please let us know at efyeditorial@efyindia.com

We would love to hear your good ideas too!

electronics
FOR YOU

the printing process.

When the design to be printed is created and fed to the printer software, most of it shows an estimate of the material consumed for that particular project. This will give the user an idea of the cost to produce a particular model. Nowadays, a kilogram of material is available for around ₹ 4000.

The lowest cost of parts is found in powder-based printing technologies. Inexpensive gypsum is used as base material and the unused powder is continuously recycled and reused by the device. This results in the reduction of cost by one-third or even one-half compared to other technologies.

Usability, help and support. The model should be user-friendly. Check whether the company offers support by mail, telephone, live chat, etc. Many companies do not provide warranty for the printer. So it is safe to double-check that the product you are buying has a warranty cover or not.

SD card and display support are some other features one can look for in a 3D printer model. Machines that offer these features can work without being connected to a PC.

Challenges and constraints. A huge limitation in the 3D printing technology is that each object can be printed using only one material. It is not possible to use different types of materials at the same time to create a model. Printing in metal is also relatively new in this field. But extensive research is going on in this field and we should soon see mixed materials and fully assembled products coming out of the printer. "In the future, we are going to start to see mixed materials, electronics 3D printing and conductive materials as well," says CarineCarmy, director of marketing, Shapeways 3D Printing Marketplace and Community.

Another hurdle is to increase the efficiency of the printing software. A smarter software could give real-time feedback of the project to be printed,

optimise design taking into consideration the physical properties required for the product and the properties of material to be used. A lot of investment is being made in this field.

Our Facebook survey suggested that price of the machine is the major issue from a user's point of view at present. A 3D printer with good specifications can be pretty expensive. One respondent from a professional design background commented on our survey, "3D printers are an expensive fad. For routine mechanical prototype makers, pay off could be sooner, but for occasional mixed technology makers it still does not make sense of buying one." Alan Campbell, a member of our Facebook page EFY's Electronics Design Community, commented that he would not prefer buying a 3D printer, "but if someone sets up a (local) shop printing parts made to order, I would give them my business."

For people like Alan, who are not ready to buy their own 3D printer, there are services to get access to high-end industrial 3D printer machines. Such professional 3D printing services could offer high-quality prototypes in about 30 different materials, at affordable costs.

According to CarineCarmy, the home printer is still fantastic for prototyping, but not yet for end goods. There are several 3D printing service providers present today; Shapeways, 3Dgiprints, Clarity 3D, 3D Hubs, KadKraft Systems, C4L, Netgains, Sculpteo, i.materialise, Ponoko, 3Ding and CADD Centre to name a few.

3D printing is improving the manufacturing and production of goods worldwide. It could create fast and easy prototypes at a decent cost. This technology makes it possible for you to hold your ideas in your hand, to create something that never existed before; imagination is the limit. 3D printing is going to revolutionise the world. Get ready for it! ●

The author is a technical correspondent at EFY, Bengaluru